Robust Strictly Positive Real Synthesis for Fifth-Order Interval Polynomial Families

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Abstract

Every Hurwitz-stable interval family of monic real polynomials of degree five admits a single real numerator of degree five that makes all the associated transfer functions strictly positive real. We prove this existence theorem. The complete proof has also been formalized in Lean 4.

Keywords: robust strictly positive real synthesis; interval polynomial; Hurwitz stability; Kharitonov polynomial; root interlacing; convex separation; Lean 4

1 Introduction

Strictly positive real transfer functions arise in passivity, absolute stability, adaptive control, and system identification. Dasgupta and Bhagwat [1] gave early design conditions for adaptive output-error identification; Yu and Wang [2] survey robust SPR analysis and synthesis. Under coefficient uncertainty, synthesis requires a single numerator for an entire family of denominators. The numerator may depend on the family, but it must remain fixed as the denominator coefficients and the frequency vary.

For a real interval polynomial family, Kharitonov's theorem [3] reduces Hurwitz stability to four distinguished vertex polynomials. For any prescribed numerator, the SPR inequalities also reduce to these four vertices. The existence problem is harder: four separate numerators, each suitable for one vertex, do not supply one numerator suitable for all four. The issue is therefore the simultaneous feasibility of the frequency inequalities, rather than the number of denominators at which a proposed solution must be tested.

The fourth-order literature illustrates this distinction. Anderson et al. [4] used a linear-programming formulation to claim SPR synthesis for every stable fourth-order interval family. Betser and Zeheb [5] exhibited a synthesizable family for which that linear program was infeasible, showing that the proposed condition was not necessary. Their example did not disprove the existence claim, which Yu and Wang [6] subsequently established. Related low-order results [7] treated stable segments and interval families of order at most four and developed the four-vertex reduction.

For polynomial segments, synthesis was established at order five [8] and order six [9], and then at arbitrary order by Yu, Wang, and Ackermann [10], with a further treatment in [11]. These results concern a single scalar interpolation parameter. An interval family allows all independent coefficient combinations in a box, and numerators obtained for its individual segments need not coincide. The arbitrary-order segment theorem therefore leaves the interval existence problem unresolved.

We prove that every Hurwitz-stable interval family K of monic real quintic polynomials has a common real numerator b of actual degree five for which $\mathrm{Re}(b(i\omega)/a(i\omega)) > 0$ for every

$a \in K$ and every $\omega \in \mathbb{R}$ (Theorem 2.1). The argument exploits both the independent coefficient intervals and the quadratic degrees of the even and odd components.

The proof begins with the four-vertex reduction and a compact convex set in the six-dimensional numerator-coefficient space. A strict separating vector is a common quintic numerator. If synthesis fails, continuous expansion from a singleton family yields a critical family with a nonzero limiting vector and nonnegative pairings. The constraint vectors at which the limiting pairing vanishes lie in a hyperplane, so their positive weighted dependence uses at most six support points (Proposition 3.3). This finite representation is accompanied by inequalities that remain valid for every denominator and every frequency.

Independent coefficient variations first force the numerator components to be coprime. The weighted identities then become an interpolation identity whose degree bounds allow only a constant proportionality factor (Proposition 4.6). The resulting alternating signs, the common order of the denominator root intervals, and contact multiplicities reduce the possible configurations to five (Proposition 5.5). This reduction includes an explicit finite enumeration, recorded in full in Appendix A. Three comparisons of value ratios at vertices with a shared component exclude all five configurations. The final proof combines these propositions to obtain strict positivity.

Sections 2–6 develop this argument in the order just described. The proof and its complete mathematical dependency chain have also been formalized in Lean 4; Section 8 describes the separate development and gives its source location. The mathematical proof presented here is self-contained. Its support and degree bounds are specific to the quintic problem, and no higher-order interval result is asserted.

2 Main result and preliminaries

2.1 Main theorem

A real polynomial is Hurwitz stable if all its zeros lie in the open left half-plane. A polynomial family is stable if every member is stable. For real polynomials a and b of the same degree, the rational function b/a is strictly positive real (SPR) if a is Hurwitz stable and

$$\operatorname{Re} \frac{b(i\omega)}{a(i\omega)} > 0 \qquad (\omega \in \mathbb{R}).$$

Theorem 2.1. Let $l_j, u_j \in \mathbb{R}$ satisfy $l_j \le u_j$ for $1 \le j \le 5$, and define

$$K = \left\{ s^5 + a_1 s^4 + a_2 s^3 + a_3 s^2 + a_4 s + a_5 : l_j \le a_j \le u_j,\ 1 \le j \le 5 \right\}. \tag{2.1}$$

If every $a \in K$ is Hurwitz stable, then there is a single polynomial $b \in \mathbb{R}[s]$ with $\deg b = 5$ such that

$$\operatorname{Re} \frac{b(i\omega)}{a(i\omega)} > 0 \qquad \text{for all } a \in K \text{ and } \omega \in \mathbb{R}. \tag{2.2}$$

Throughout the proof, K denotes a stable family of the form (2.1). Its coefficient endpoints, four vertex polynomials, and component polynomials always refer to the same family.

2.2 Real polynomial components and four vertex polynomials

For $a(s) = s^5 + a_1 s^4 + \cdots + a_5 \in K$, put $t = \omega^2$. Substituting $s = i\omega$ gives

$$a(i\omega) = E_a(t) + i\omega O_a(t), \tag{2.3}$$

where $E_a, O_a \in \mathbb{R}[t]$ are defined by

$$E_a(t) = a_1 t^2 - a_3 t + a_5, \quad O_a(t) = t^2 - a_2 t + a_4. \tag{2.4}$$

We call E_a, O_a the even and odd components; they are polynomials in t, with $a(s) = E_a(-s^2) + sO_a(-s^2)$.

Write a candidate numerator as

$$b(s) = b_0 s^5 + b_1 s^4 + b_2 s^3 + b_3 s^2 + b_4 s + b_5,$$

and define

$$E_b(t) = b_1 t^2 - b_3 t + b_5, \quad O_b(t) = b_0 t^2 - b_2 t + b_4. \tag{2.5}$$

The intermediate arguments allow $b_0 = 0$, so the numerator is not yet required to have degree five. Define the real pairing polynomial

$$P_{a,b}(t) = E_b(t)E_a(t) + tO_b(t)O_a(t). \tag{2.6}$$

Stability gives $a(i\omega) \neq 0$. Hence

$$\operatorname{Re} \frac{b(i\omega)}{a(i\omega)} = \frac{P_{a,b}(t)}{|a(i\omega)|^2}. \tag{2.7}$$

The real part is therefore positive at every real frequency if and only if $P_{a,b}(t) > 0$ for all $t \geq 0$.

To describe variation of the denominator coefficients, write

$$E_-(t) = l_1 t^2 - u_3 t + l_5, \quad E_+(t) = u_1 t^2 - l_3 t + u_5 \tag{2.8}$$

$$O_-(t) = t^2 - u_2 t + l_4, \quad O_+(t) = t^2 - l_2 t + u_4. \tag{2.9}$$

For each fixed $t \geq 0$, the values $(E_a(t), O_a(t))$ fill exactly the rectangle

$$\mathcal{R}_t = \{(E, O) \in \mathbb{R}^2 : E_-(t) \leq E \leq E_+(t),\ O_-(t) \leq O \leq O_+(t)\}.$$

This follows from the independent variation of the even-indexed and odd-indexed coefficient groups and the signs of their coefficients for $t \geq 0$.

Define four fixed quintic polynomials by

$$\begin{cases} A(s) = s^5 + u_1 s^4 + u_2 s^3 + l_3 s^2 + l_4 s + u_5, \\ B(s) = s^5 + u_1 s^4 + l_2 s^3 + l_3 s^2 + u_4 s + u_5, \\ C(s) = s^5 + l_1 s^4 + l_2 s^3 + u_3 s^2 + u_4 s + l_5, \\ D(s) = s^5 + l_1 s^4 + u_2 s^3 + u_3 s^2 + l_4 s + l_5. \end{cases} \tag{2.10}$$

These are the four Kharitonov polynomials of K, hereafter called the vertex polynomials. Their components are

Vertex J	(E_J, O_J)	Vertex J	(E_J, O_J)
A	(E_+, O_-)	C	(E_-, O_+)
B	(E_+, O_+)	D	(E_-, O_-)

Their component values are the four vertices of $\mathcal{R}_t$.

Lemma 2.2 (Four-vertex reduction)**.** Fix a real polynomial b of degree at most five. Then $P_{a,b}(t) > 0$ for all $a \in K$ and $t \geq 0$ if and only if $P_{J,b}(t) > 0$ for all $J \in \{A, B, C, D\}$ and $t \geq 0$. The statement remains valid with every $>$ replaced by $\geq$.

Proof. For fixed $t \geq 0$, the linear function $E_b(t)E + tO_b(t)O$ attains its minimum on $\mathcal{R}_t$ at a vertex. The same argument proves both assertions. □

2.3 Root interlacing for stable polynomials

The Hermite–Biehler theorem relates Hurwitz stability to interlacing of the zeros of the even and odd parts. We state only the quintic case needed here; see Holtz [12].

Lemma 2.3. Let a be a monic real quintic polynomial that is Hurwitz stable. Then $a_j > 0$ for $1 \le j \le 5$, and both E_a and O_a have two simple positive roots. If the roots of E_a are $\alpha_1 < \alpha_2$ and those of O_a are $\beta_1 < \beta_2$, then

$$0 < \alpha_1 < \beta_1 < \alpha_2 < \beta_2. \tag{2.11}$$

Consequently,

$$E_a(t) = a_1(t-\alpha_1)(t-\alpha_2), \quad O_a(t) = (t-\beta_1)(t-\beta_2).$$

The usual frequency-variable formulation applies to $E_a(\omega^2)$ and $\omega O_a(\omega^2)$; squaring their positive roots gives (2.11). The coefficient signs follow from the real linear and quadratic factors of a Hurwitz polynomial.

2.4 Root bounds for the entire interval family

Stability of the four vertex polynomials implies that each of E_-, E_+, O_-, O_+ has two simple positive roots. Label them in increasing order within each component by

$$E_-(t) = l_1(t-e_L)(t-f_U), \quad E_+(t) = u_1(t-e_U)(t-f_L) \tag{2.12}$$

$$O_-(t) = (t-o_L)(t-p_U), \quad O_+(t) = (t-o_U)(t-p_L). \tag{2.13}$$

Proposition 2.4 (Uniform root bounds)**.** If $l_j < u_j$ for $1 \le j \le 5$, the endpoints above satisfy

$$0 < e_L < e_U < o_L < o_U < f_L < f_U < p_L < p_U. \tag{2.14}$$

For every $a \in K$, the two roots of E_a lie in $[e_L, e_U]$ and $[f_L, f_U]$, respectively, and those of O_a lie in $[o_L, o_U]$ and $[p_L, p_U]$.

Proof. Put $\Delta_j = u_j - l_j > 0$. The component widths are

$$E_+(t) - E_-(t) = \Delta_1 t^2 + \Delta_3 t + \Delta_5 > 0, \tag{2.15}$$

$$O_+(t) - O_-(t) = \Delta_2 t + \Delta_4 > 0 \qquad (t \ge 0). \tag{2.16}$$

At either root of E_+, we have $E_- < 0$. Since $l_1 > 0$, the polynomial E_- is negative only between its own two roots. Thus $e_L < e_U < f_L < f_U$. Similarly, both roots of O_+ lie between those of O_-, giving $o_L < o_U < p_L < p_U$. Applying Lemma 2.3 to A, B, C, respectively, also gives

$$e_U < o_L, \quad o_U < f_L, \quad f_U < p_L.$$

These inequalities give (2.14). For the even component, the inequalities $E_- \le E_a \le E_+$ imply

$$E_a(e_L) \ge 0, \quad E_a(e_U) \le 0,$$
$$E_a(f_L) \le 0, \quad E_a(f_U) \ge 0.$$

The intermediate value theorem places one root in each interval. The quadratic degree excludes any others. The same argument applies to $O_- \le O_a \le O_+$. □

Number the four closed root intervals and the gaps between and outside them consecutively as $0, \dots, 8$, as in Table 1. Signs are listed in the order $(E_a(t), O_a(t))$, at points where neither component vanishes.

The closed root intervals include their endpoints; the intervening gaps do not. Thus the region index is nondecreasing in t, independently of which $a \in K$ supplies the signs.

Table 1: Possible signs of the denominator components in each region

Region and range	Signs	Region and range	Signs
0 $[0, e_L)$	$++$	5 $[f_L, f_U]$	$--, +-$
1 $[e_L, e_U]$	$++, -+$	6 (f_U, p_L)	$+-$
2 (e_U, o_L)	$-+$	7 $[p_L, p_U]$	$+-, ++$
3 $[o_L, o_U]$	$-+, --$	8 (p_U, ∞)	$++$
4 (o_U, f_L)	$--$		

3 A critical interval family and finite weighted identities

A separating vector gives a common numerator. At the first parameter where separation fails, a nonzero limiting vector confines every support point to one hyperplane; this improves the usual seven-point convex-hull bound to six.

3.1 A vector representation of the frequency constraints

Let $\mathbf{b} = (b_0, \dots, b_5) \in \mathbb{R}^6$ be the coefficient vector of b, and define

$$F(a,t) = (t^3 O_a(t), t^2 E_a(t), -t^2 O_a(t), -t E_a(t), t O_a(t), E_a(t)).$$

Then $\mathbf{b} \cdot F(a,t) = P_{a,b}(t)$.

To include arbitrarily large t, use the bijective change of variables

$$x = \frac{t}{1+t} \in [0,1), \quad t = \frac{x}{1-x},$$

and multiply F by the positive factor $(1+t)^{-5}$ to define

$$G(a,x) = (1-x)^5 F\left(a, \frac{x}{1-x}\right) \qquad (0 \le x < 1).$$

Since $\deg_t F \le 5$, the components of G extend polynomially to $x = 1$, where

$$G(a,1) = v_\infty = (1,0,0,0,0,0). \tag{3.1}$$

Thus $x = 1$ represents the limit as $t \to \infty$, and $\mathbf{b} \cdot v_\infty = b_0$ records the coefficient of the quintic term of the numerator.

Collect the four curves and their endpoints into

$$S_K = \{G(J,x) : J \in \{A,B,C,D\},\ 0 \le x \le 1\}, \quad C_K = \operatorname{conv} S_K.$$

The set S_K is compact. Carathéodory's theorem [13] represents each point of C_K by at most seven points of S_K; the resulting compact parameter space also proves that C_K is compact.

Lemma 3.1. If $0 \notin C_K$, there exists a real polynomial b of degree five satisfying (2.2).

Proof. By compactness, C_K has a point $z = (z_0, \dots, z_5)$ of minimum norm, with $\|z\| > 0$. For any $y \in C_K$, convexity gives $z + \tau(y - z) \in C_K$ for $0 \le \tau \le 1$. The right derivative of its squared norm at $\tau = 0$ is nonnegative, so

$$2z \cdot (y - z) \ge 0, \quad z \cdot y \ge \|z\|^2 > 0.$$

Take $b(s) = z_0 s^5 + z_1 s^4 + \cdots + z_5$. Setting $y = v_\infty$ gives $z_0 > 0$, and hence $\deg b = 5$. Taking $y = G(J, t/(1+t))$ and multiplying by $(1+t)^5$ gives $P_{J,b}(t) > 0$ for all four vertex polynomials and all $t \ge 0$. Lemma 2.2 and (2.7) complete the proof. □

3.2 Interval expansion and the critical parameter

Lemma 3.2. Every stable interval family is contained in a stable interval family with all coefficient intervals of positive length.

Proof. By continuous dependence of polynomial roots on the coefficients, the monic Hurwitz quintics form an open subset of the coefficient space $\mathbb{R}^5$. The coefficient set of K is a compact subset of this open set and therefore has positive distance from the unstable coefficient set. For sufficiently small $\varepsilon > 0$, enlarge each $[l_j, u_j]$ to $[l_j - \varepsilon, u_j + \varepsilon]$. Every point in the enlarged box has Euclidean distance at most $\sqrt{5}\,\varepsilon$ from the original coefficient set. It therefore still represents a stable polynomial, and every interval now has positive length. □

A common numerator for an enlarged family also works for every subfamily. It therefore suffices to consider $l_j < u_j$. Set $\Delta_j = u_j - l_j$ and define the nested stable families

$$K_\theta = \left\{ s^5 + a_1 s^4 + \cdots + a_5 : l_j \le a_j \le l_j + \theta\Delta_j,\ 1 \le j \le 5 \right\}, \qquad 0 \le \theta \le 1.$$

Here $K_0 = \{a^-\}$, where $a^-(s) = s^5 + l_1 s^4 + \cdots + l_5$, and $K_1 = K$.

Proposition 3.3 (Critical family and finite support)**.** Suppose $l_j < u_j$ for $1 \le j \le 5$ and K admits no real polynomial of degree five satisfying (2.2). There exist $\theta_* \in (0, 1]$, a polynomial $b \in \mathbb{R}[s]$ of degree at most five with coefficient vector $\mathbf{b} \in \mathbb{R}^6$ of norm one and $b_0 \ge 0$, and finite data

$$0 \le t_1 < \cdots < t_m, \quad a_i \in K_{\theta_*}, \quad \mu_i > 0\ (1 \le i \le m), \quad \mu_\infty \ge 0,$$

with $m \ge 1$ and $m + \mathbf{1}_{\{\mu_\infty > 0\}} \le 6$, such that $0 \in C_{K_{\theta_*}}$ and

$$P_{a,b}(t) \ge 0 \qquad (a \in K_{\theta_*},\ t \ge 0). \tag{3.2}$$

The finite data satisfy

$$\sum_{i=1}^{m} \mu_i F(a_i, t_i) + \mu_\infty v_\infty = 0, \tag{3.3}$$

and

$$P_{a_i,b}(t_i) = 0 \quad (1 \le i \le m), \qquad \mu_\infty b_0 = 0. \tag{3.4}$$

Moreover, $0 \notin C_{K_\theta}$ for $0 \le \theta < \theta_*$. The family K_{θ_*} is stable and every coefficient interval has positive length.

Proof. Put $C_\theta = C_{K_\theta}$ and write $Z = \{\theta \in [0, 1] : 0 \in C_\theta\}$. The coefficients of the four vertex polynomials and the map G depend continuously on θ. Using convex representations with at most seven terms, and letting θ, the seven curve parameters, and the seven weights vary over their respective compact sets, we see that $\{(\theta, y) : 0 \le \theta \le 1,\ y \in C_\theta\}$ is compact. Thus Z is closed. If K has no common numerator, Lemma 3.1 gives $1 \in Z$. At $\theta = 0$, however, the choice $b = a^-$ satisfies $P_{a^-,a^-}(t) = |a^-(i\sqrt{t})|^2 > 0$, and the inner product at infinity is one. This vector therefore has positive inner product with every point of S_{K_0} and hence of C_0, so $0 \notin Z$. Since Z is a nonempty closed subset of $[0, 1]$ and excludes zero, its minimum θ_* exists and is strictly positive. For every $\theta < \theta_*$, $0 \notin C_\theta$.

Choose $\theta_n \uparrow \theta_*$ with $\theta_n < \theta_*$. Each C_{θ_n} excludes the origin. Take a separating vector from Lemma 3.1 and normalize it, writing the result as $\mathbf{b}^{(n)}$. Compactness of the unit sphere gives a convergent subsequence with a unit limit $\mathbf{b}$. Let b be the polynomial with this coefficient vector. For a fixed vertex label $J \in \{A, B, C, D\}$, let J_θ be the corresponding vertex polynomial of K_θ. For each fixed $x \in [0, 1]$,

$$\mathbf{b} \cdot G(J_{\theta_*}, x) = \lim_{n \to \infty} \mathbf{b}^{(n)} \cdot G(J_{\theta_n}, x) \ge 0.$$

The same convergent subsequence is used for every fixed J, x. Continuity therefore gives nonnegativity for all four curves and every $x \in [0,1]$. At $x = 1$ this yields $b_0 \geq 0$; Lemma 2.2 gives (3.2). Normalization before taking the limit is essential: $\|\mathbf{b}\| = 1$, so the limiting polynomial is nonzero even though strict positivity may be lost. For the remainder of this proof, write $K_* = K_{\theta_*}$.

Since $0 \in C_{K_*}$, the origin is a convex combination of finitely many points of S_{K_*}. Choose such a representation with the fewest points. After removing zero weights, there are $v_i \in S_{K_*}$ and $\lambda_i > 0$, with $\sum_i \lambda_i = 1$, such that

$$0 = \sum_{i=1}^{n} \lambda_i v_i.$$

These points are affinely independent. Otherwise, there would be real numbers c_i, not all zero, with $\sum_i c_i = 0$ and $\sum_i c_i v_i = 0$. Replacing the weights by $\lambda_i - \tau c_i$ with $\tau = \min_{c_i > 0} \lambda_i / c_i$ preserves nonnegativity, their sum of one, and the represented point, but removes at least one nonzero term. This contradicts minimality.

By (3.2) and $b_0 \geq 0$, each $\mathbf{b} \cdot v_i \geq 0$. Their positively weighted sum is zero, so every inner product is zero. Since $\mathbf{b} \neq 0$, the points lie in the five-dimensional hyperplane $\mathbf{b}^{\perp} = \{v \in \mathbb{R}^6 : \mathbf{b} \cdot v = 0\}$. A five-dimensional affine space contains at most six affinely independent points, giving $n \leq 6$. The improvement from seven to six uses the nonzero limiting vector, not a reduction in the original coefficient-space dimension.

Each support point at a finite frequency has the form $G(a_i, x_i)$, where a_i is a vertex polynomial and $x_i < 1$. Put

$$t_i = \frac{x_i}{1 - x_i}, \quad \mu_i = \frac{\lambda_i}{(1 + t_i)^5} > 0,$$

Then $\lambda_i G(a_i, x_i) = \mu_i F(a_i, t_i)$. Denote the total weight at $x = 1$ by $\mu_\infty \geq 0$. This gives (3.3). The positive rescaling changes the weights but preserves the vector sum and each zero inner product; the new weights need not sum to one.

If two finite terms have the same parameter t, denominators a, a', and weights μ, μ', put $\bar{\mu} = \mu + \mu'$ and form the coefficientwise combination $\bar{a} = (\mu a + \mu' a')/\bar{\mu}$. Convexity of the coefficient intervals gives $\bar{a} \in K_*$. Since $F(a, t)$ is affine in the denominator coefficients,

$$\mu F(a, t) + \mu' F(a', t) = \bar{\mu} F(\bar{a}, t).$$

The merged coefficient vector remains monic and in the same interval box, hence is Hurwitz stable. Its two component values and its vector F are the corresponding weighted averages. Since both original pairings vanish, the merged pairing vanishes as well. Nonnegativity for the merged denominator follows from (3.2), which already holds throughout K_*. Merging does not increase the number of terms and does not alter the weight at infinity. Repetition gives distinct finite nodes with positive weights and the asserted support bound. There must be a finite node, since a positive multiple of v_∞ alone cannot sum to zero. The zero inner products prove (3.4). Finally, $\theta_* > 0$ makes every interval of K_* have length $\theta_* \Delta_j > 0$. □

3.3 Moment form of the finite representation

From now until the final proof, fix data supplied by Proposition 3.3 and denote its critical family again by K, with endpoints l_j, u_j. Thus $\Delta_j > 0$, $b \neq 0$, $b_0 \geq 0$, and $P_{a,b} \geq 0$ on $[0, \infty)$ for every $a \in K$. A zero of $P_{a,b}$ on this half-line is called a contact; its parameter is $t = \omega^2$.

Write $E_i = E_{a_i}(t_i)$ and $O_i = O_{a_i}(t_i)$. Expanding (3.3) in the six coordinates of F gives

$$\sum_i \mu_i t_i^j E_i = 0 \qquad (j = 0, 1, 2), \tag{3.5}$$

$$\sum_i \mu_i t_i O_i = 0, \tag{3.6}$$

$$\sum_i \mu_i t_i^2 O_i = 0, \tag{3.7}$$

$$\sum_i \mu_i t_i^3 O_i = -\mu_\infty. \tag{3.8}$$

Only the first coordinate of (3.3) receives a contribution from infinity. The finite denominators belong to K, although merging may have moved them away from the vertices.

4 Algebraic consequences of nonnegativity and interpolation

Retain the critical data fixed above. Nonnegativity for the entire box permits independent coefficient perturbations at a contact. These perturbations exclude common roots of E_b, O_b; the resulting coprimality is the essential input to the interpolation argument.

4.1 Exclusion of common roots

Lemma 4.1. Let a be a monic Hurwitz-stable real quintic polynomial, and let $c, d \in \mathbb{R}$. If $cE_a(t) + tdO_a(t) \geq 0$ for every $t \geq 0$, then $c = d = 0$.

Proof. At the roots α_1, α_2 of Lemma 2.3, the signs of O_a are positive and negative, respectively. Thus $d \geq 0$ and $d \leq 0$, giving $d = 0$. At $t = 0$, the inequality $E_a(0) = a_5 > 0$ gives $c \geq 0$, while $E_a(\beta_1) < 0$ gives $c \leq 0$. □

Lemma 4.2. The polynomials E_b and O_b have no common positive root.

Proof. Suppose $E_b(r) = O_b(r) = 0$ for some $r > 0$. Then $P_{a,b}(r) = 0$ for every $a \in K$. Since $P_{a,b}$ is nonnegative on $[0, \infty)$, it has a local minimum at r, so $P'_{a,b}(r) = 0$. The derivative equation simplifies to

$$E'_b(r)E_a(r) + rO'_b(r)O_a(r) = 0.$$

Changing one coefficient independently leaves the denominator in K, so nonnegativity on the whole half-line is preserved. Keep the other coefficients fixed and change a_5 from its lower to its upper endpoint. Subtracting the two equations gives $\Delta_5 E'_b(r) = 0$, hence $E'_b(r) = 0$. Independently changing a_4 gives $r\Delta_4 O'_b(r) = 0$, hence $O'_b(r) = 0$. Since E_b, O_b have degree at most two, simultaneous vanishing of their values and derivatives implies

$$E_b(t) = b_1(t - r)^2, \quad O_b(t) = b_0(t - r)^2.$$

Therefore,

$$P_{a,b}(t) = (t - r)^2\{b_1 E_a(t) + tb_0 O_a(t)\} \geq 0.$$

Divide by the positive square when $t \neq r$ and use continuity at $t = r$. The expression in braces is nonnegative on the half-line. Lemma 4.1 gives $b_1 = b_0 = 0$, and the two factorizations then force the other four coefficients to vanish, contradicting $b \neq 0$. □

Lemma 4.3. Let $a \in K$. If $r > 0$ and $P_{a,b}(r) = 0$, then $E_b(r), O_b(r), E_a(r), O_a(r)$ are all nonzero.

Proof. Suppose $E_b(r) = 0$. By Lemma 4.2, $O_b(r) \neq 0$, so the contact equation forces $O_a(r) = 0$. Changing a_5 preserves this contact. Subtracting the derivative equations at the two coefficient endpoints again gives $E_b'(r) = 0$. The original derivative equation reduces to $rO_b(r)O_a'(r) = 0$, contradicting simplicity of the root of O_a.

Thus $E_b(r) \neq 0$. If $O_b(r) = 0$, then $E_a(r) = 0$. Changing a_4 independently and subtracting the derivative equations gives $r\Delta_4 O_b'(r) = 0$. The original equation then gives $E_b(r)E_a'(r) = 0$, again contradicting simplicity. Finally, if either denominator component vanishes, the contact equation forces the other to vanish as well. This would imply $a(i\sqrt{r}) = 0$, contrary to stability. □

4.2 Coefficient signs and root ordering

Proposition 4.4 (Numerator components)**.** The critical polynomial satisfies $b_5 \geq 0$ and

$$b_1, b_2, b_3, b_4 > 0. \tag{4.1}$$

If $b_0 > 0$, there are roots such that

$$0 \leq e < k < f < \ell, \quad E_b = b_1(t-e)(t-f), \quad O_b = b_0(t-k)(t-\ell). \tag{4.2}$$

If $b_0 = 0$, then $\deg b = 4$ and

$$0 \leq e < k < f, \quad E_b = b_1(t-e)(t-f), \quad O_b = -b_2(t-k). \tag{4.3}$$

In both cases E_b, O_b are coprime in $\mathbb{R}[t]$, and $b_5 = 0$ if and only if $e = 0$.

Proof. Take any $a \in K$ and use the root notation of Lemma 2.3. At a root of E_a, the expression in (2.6) reduces to tO_bO_a; at a root of O_a, it reduces to E_bE_a. Nonnegativity of $P_{a,b}$ and Lemma 4.3 therefore give

$$O_b(\alpha_1) > 0, \quad O_b(\alpha_2) < 0, \quad E_b(\beta_1) < 0, \quad E_b(\beta_2) > 0. \tag{4.4}$$

At these denominator roots, equality would violate Lemma 4.3. Also $P_{a,b}(0) = a_5 b_5 \geq 0$ gives $b_5 \geq 0$.

For $q(t) = ut^2 - vt + w$ and $0 < x < y$, the identity

$$xq(y) - yq(x) = (y-x)(uxy - w) \tag{4.5}$$

allows us to determine the coefficient signs directly. For $q = E_b$, take $x = \beta_1$ and $y = \beta_2$. The left-hand side is strictly positive, so $b_5 \geq 0$ implies $b_1 > 0$. For $q = O_b$, take $x = \alpha_1$ and $y = \alpha_2$. The left-hand side is strictly negative, so $b_0 \geq 0$ implies $b_4 > 0$. The inequality $O_b(\alpha_2) - O_b(\alpha_1) < 0$ then gives $b_2 > 0$. If $b_3 \leq 0$, the value $E_b(\beta_1)$ would be a sum of nonnegative terms, contrary to (4.4). This proves (4.1).

The inequalities $E_b(0) \geq 0$ and $E_b(\beta_1) < 0 < E_b(\beta_2)$ give roots satisfying

$$0 \leq e < \beta_1 < f < \beta_2, \quad E_b(t) = b_1(t-e)(t-f).$$

If $b_0 > 0$, the inequalities $O_b(\alpha_1) > 0 > O_b(\alpha_2)$ and the positive leading coefficient give $\alpha_1 < k < \alpha_2 < \ell$ with $O_b(t) = b_0(t-k)(t-\ell)$. When $e > 0$, we have $O_a(e) > 0$, so nonnegativity of $P_{a,b}$ and the absence of common positive numerator roots imply $O_b(e) > 0$. When $e = 0$, the same conclusion follows from $O_b(0) = b_4 > 0$. At f, we have $O_a(f) < 0$ and similarly obtain $O_b(f) < 0$. Denominator interlacing and the preceding bounds also give $e < \beta_1 < \alpha_2 < \ell$. Together with $O_b(e) > 0$ and $O_b(f) < 0$, this gives (4.2). If $b_0 = 0$, then $O_b = -b_2(t-k)$ with $k = b_4/b_2 > 0$; the same signs yield (4.3). The displayed roots exhaust both components and are disjoint, proving coprimality. Finally, $b_5 = b_1 ef$ and $b_1, f > 0$ give the assertion about e. □

4.3 Interpolation identities from vanishing moments

The moment identities impose low degrees on interpolation polynomials, while the contacts force one polynomial identity. For m distinct finite nodes, write

$$L(t) = \prod_{i=1}^{m}(t - t_i), \quad \eta_i = \frac{1}{L'(t_i)} = \frac{1}{\prod_{j\neq i}(t_i - t_j)},$$
$$z_i = \mu_i E_i, \quad y_i = \mu_i t_i O_i.$$

For any weight vector $v = (v_1, \dots, v_m)$, define

$$R_v(t) = \sum_{i=1}^{m} v_i \prod_{j\neq i}(t - t_j).$$

Take $U = R_z$ and $V = R_y$, where $z = (z_1, \dots, z_m)$ and $y = (y_1, \dots, y_m)$. At each node,

$$U(t_i) = \frac{z_i}{\eta_i}, \quad V(t_i) = \frac{y_i}{\eta_i}. \tag{4.6}$$

Lemma 4.5 (Vanishing moments and a degree bound)**.** Let $m \geq 1$, let $t_1, \dots, t_m \in \mathbb{R}$ be distinct, and let $z_1, \dots, z_m \in \mathbb{R}$. For an integer $k \geq 0$, if $\sum_i t_i^j z_i = 0$ for $0 \leq j < k$, then $\deg R_z \leq m - k - 1$, with $\deg 0 = -\infty$.

Proof. For $k = 0$, the bound follows from the definition of R_z. Suppose $k \geq 1$. Using $t = (t - t_i) + t_i$ in each term of tR_z gives

$$tR_z = R_{(t_i z_i)_i} + \Big(\sum_i z_i\Big) L.$$

When the zeroth moment vanishes, the term containing L is zero, so $tR_z = R_{(t_i z_i)_i}$. Apply the same identity successively to the weights $(t_i^j z_i)_i$ for $j = 1, \dots, k-1$. Their zeroth moments are precisely the original moments of orders one through $k - 1$, so the L term vanishes at every step. Hence

$$t^k R_z = R_{(t_i^k z_i)_i}.$$

The right-hand side has degree at most $m - 1$. If $R_z \neq 0$, the left-hand side has degree $k + \deg R_z$, proving the bound. □

Proposition 4.6 (Interpolation rigidity)**.** For the critical data and the polynomials $U = R_z$, $V = R_y$ above, there is a real constant $c \neq 0$ such that

$$U = cO_b, \quad V = -cE_b. \tag{4.7}$$

Consequently, for every $1 \leq i \leq m$,

$$\mu_i E_i = c\eta_i O_b(t_i), \quad \mu_i t_i O_i = -c\eta_i E_b(t_i). \tag{4.8}$$

Proof. The zeroth, first, and second moments of z vanish. The zeroth and first moments of y vanish, and its second moment equals $-\mu_\infty$. Applying Lemma 4.5 to z, y gives

$$\deg U \leq m - 4, \tag{4.9}$$
$$\deg V \leq m - 3, \tag{4.10}$$
$$\mu_\infty = 0 \implies \deg V \leq m - 4. \tag{4.11}$$

There is at least one finite node, and $E_i \neq 0$ at every finite node: this follows from Lemma 4.3 at positive nodes, and from $E_i = a_{i,5} > 0$ at a zero node. Thus (4.6) implies $U \neq 0$.

The contact equations and (4.6) give

$$(E_bU + O_bV)(t_i) = 0 \qquad (1 \le i \le m).$$

The degree bounds above imply $\deg(E_bU) \le m-2$ and $\deg(O_bV) \le m-1$. Their sum therefore has degree at most $m-1$. Since it vanishes at m distinct nodes, it is identically zero. Thus

$$E_bU + O_bV \equiv 0. \tag{4.12}$$

By Proposition 4.4, E_b, O_b are coprime. Hence (4.12) implies $E_b \mid V$. Write $V = E_bQ$ and cancel the nonzero polynomial E_b in (4.12) to obtain $U = -O_bQ$. If $Q = 0$, then $U = 0$, a contradiction. Hence $Q \neq 0$. Since $\deg E_b = 2$, it follows that $\deg V = 2 + \deg Q$.

If $\mu_\infty = 0$, the support bound $m \le 6$ and (4.11) give $\deg V \le 2$. If $\mu_\infty > 0$, then $m \le 5$, and (4.10) gives the same bound. Thus Q is a nonzero constant. Writing $Q = -c$ proves (4.7), and evaluation at the nodes using (4.6) proves (4.8). □

Proposition 4.7 (Support size)**.** The critical data satisfy exactly one of the following two alternatives, with c as in Proposition 4.6:

(i) $m = 5$, $\mu_\infty > 0$, $b_0 = 0$, $c > 0$;

(ii) $m = 6$, $\mu_\infty = 0$.

Proof. Since $\deg V = 2$, when $\mu_\infty = 0$, equation (4.11) forces $m \ge 6$, and hence $m = 6$. When $\mu_\infty > 0$, equation (4.10) forces $m \ge 5$, and hence $m = 5$; the contact equation at infinity gives $b_0 = 0$. In this case, comparing the quartic coefficients in $t^2V = R_{(t_i^2 y_i)_i}$ yields

$$[t^2]V = \sum_i t_i^2 y_i = -\mu_\infty.$$

On the other hand, equation (4.7) gives $[t^2]V = -cb_1$. Since $b_1 > 0$, we obtain $c = \mu_\infty / b_1 > 0$. □

5 Contact multiplicities and configurations

Retain the critical data and write $P_J = P_{J,b}$ for $J \in \{A, B, C, D\}$. Propositions 4.4 and 4.6 give the component root order and node signs. Together with contact multiplicities, these restrict the possible ordered nodes.

5.1 The vertex associated with a contact

At a positive contact t_i, Lemma 4.3 makes both coefficients of the linear function $E_b(t_i)E + t_iO_b(t_i)O$ nonzero. The minimizing vertex is therefore uniquely determined by the numerator signs: $++, -+, --, +-$ select D, A, B, C, respectively. If the original denominator a_i has a contact at $t_i > 0$, the minimizing vertex J satisfies

$$0 \le P_{J,b}(t_i) \le P_{a_i,b}(t_i) = 0.$$

The contributions from displacing either rectangle coordinate from its minimizing endpoint are nonnegative. Their sum is zero, so both coordinate differences vanish. Hence

$$P_{J,b}(t_i) = 0, \quad E_J(t_i) = E_i, \quad O_J(t_i) = O_i. \tag{5.1}$$

We may therefore replace the denominator in (4.8) at each positive node by its corresponding vertex without changing either component value.

5.2 Multiplicity bounds and two incompatible contact patterns

Lemma 5.1. Let a be a monic Hurwitz-stable real quintic polynomial, and let $b \in \mathbb{R}[s]$ be nonzero with $\deg b \le 5$. Then $P_{a,b}$ is not the zero polynomial.

Proof. By Lemma 2.3, the polynomials E_a and tO_a are coprime: all their roots have been listed, and $E_a(0) > 0$. If $E_bE_a + tO_bO_a = 0$, then $E_a \mid O_b$. Since $\deg O_b \le 2 = \deg E_a$, we have $O_b = dE_a$ for a constant d. Canceling E_a gives $E_b = -dtO_a$. If $d \neq 0$, the right-hand side has degree three, whereas the left-hand side has degree at most two. Thus $d = 0$, so $E_b = O_b = 0$ and hence $b = 0$, a contradiction. □

Lemma 5.2 (Contact multiplicities)**.** For each vertex J, put $P = P_J$. Then the following properties hold.

(a) A vertex cannot have three distinct positive contacts.

(b) If $b_0 = b_5 = 0$, then $\deg P \le 4$ and $P(0) = 0$, so a vertex cannot have two additional positive contacts.

(c) If a vertex has two contacts $0 < r < s$, then

$$P(t) = (b_0t + d)(t - r)^2(t - s)^2, \quad d \ge 0. \tag{5.2}$$

Proof. Lemma 5.1 gives $P_J \neq 0$. Every positive zero of a nonnegative polynomial has even multiplicity, and $\deg P_J \le 5$, proving (a). When $b_0 = b_5 = 0$, the degree is at most four and zero is already a root; two positive contacts would require degree at least five, proving (b). For (c), divide by $(t - r)^2(t - s)^2$ and compare leading coefficients. The remaining factor has leading coefficient b_0, and its constant term d is nonnegative by nonnegativity on the half-line. If $b_0 = 0$, nonvanishing gives $d > 0$. □

Write $[t^j]P$ for the coefficient of t^j in P.

Lemma 5.3. There are no positive numbers $r < s < u < v$ such that A has contacts at r, s and B has contacts at u, v.

Proof. By (5.2), write

$$P_A = (b_0t + d_A)(t - r)^2(t - s)^2,$$
$$P_B = (b_0t + d_B)(t - u)^2(t - v)^2.$$

The polynomials A, B share E_+, so their pairing polynomials have the same constant term: $d_Ar^2s^2 = d_Bu^2v^2$. Since $rs < uv$, we have $d_B \le d_A$. If $b_0 = 0$, then $d_A, d_B > 0$, and the inequality is strict. Therefore,

$$[t^4](P_B - P_A) = d_B - d_A - 2b_0(u + v - r - s) < 0.$$

By the definitions of the vertices, however,

$$P_B - P_A = tO_b(O_+ - O_-) = tO_b(\Delta_2t + \Delta_4),$$

whose quartic coefficient is $b_0\Delta_2 \ge 0$, a contradiction. □

Lemma 5.4. Suppose $b_0 > 0$ and $b_5 = 0$. There are no positive numbers $r < s < u < v$ such that B has contacts at r, s and C has contacts at u, v.

Proof. Since $b_5 = 0$, we have $P_B(0) = P_C(0) = 0$. Setting $t = 0$ in (5.2) shows that $d_B = d_C = 0$. Write $q_{r,s}(t) = (t-r)(t-s)$ and $q_{u,v}(t) = (t-u)(t-v)$. Then

$$P_B = b_0 t\, q_{r,s}^2, \quad P_C = b_0 t\, q_{u,v}^2.$$

Put $\Delta E = E_+ - E_- = \Delta_1 t^2 + \Delta_3 t + \Delta_5$. Using $E_b = t(b_1 t - b_3)$ and the common component O_+, then canceling t, gives

$$b_0(q_{u,v}^2 - q_{r,s}^2) = -(b_1 t - b_3)\Delta E. \tag{5.3}$$

Set

$$S = u + v - r - s > 0, \quad \tau = \frac{uv - rs}{S} > 0, \quad W = q_{u,v} + q_{r,s}.$$

Then $q_{u,v} - q_{r,s} = -S(t - \tau)$. At $t = \tau$, the left-hand side of (5.3) vanishes, while $\Delta E(\tau) > 0$, so $b_1\tau - b_3 = 0$. Factoring the difference of squares and canceling the nonzero polynomial $t - \tau$ gives

$$b_0 SW = b_1 \Delta E.$$

The linear coefficient on the left is $-b_0 S(u + v + r + s) < 0$, whereas that on the right is $b_1\Delta_3 > 0$, a contradiction. □

5.3 The five remaining configurations

When b is quintic, equation (4.2) gives the root ordering $0 \le e < k < f < \ell$ for its components E_b, O_b. These four roots divide the positive half-line into

$$I_0 = (0, e), \quad I_1 = (e, k), \quad I_2 = (k, f), \quad I_3 = (f, \ell), \quad I_4 = (\ell, \infty).$$

These interval indices are determined by the roots of the numerator components and differ from the denominator region indices in Table 1. If $e = 0$, then I_0 is empty. In the quartic case, use I_0, I_1, I_2 and $I_3 = (f, \infty)$. Across these intervals, the signs $(\operatorname{sgn} E_b, \operatorname{sgn} O_b)$ are $++, -+, --, +-, ++$, and the contact vertices are D, A, B, C, D, respectively; omit the last entry in the quartic case. By Lemma 4.3, positive contacts cannot occur at the interval endpoints.

Proposition 5.5. For the critical data of Proposition 3.3, the finite representation has $m = 6$, $t_1 > 0$, $\mu_\infty = 0$, and $b_0, b_5 > 0$. The constant c of Proposition 4.6 is positive. If q_i is defined by $t_i \in I_{q_i}$, only the five configurations in Table 2 remain. In each configuration, the six vertices and six denominator sign pairs correspond to $t_1, \dots, t_6$ in order. The same vertex may have different signs at different nodes.

Proof. By (4.8), at positive nodes,

$$(\operatorname{sgn} E_i, \operatorname{sgn} O_i) = \operatorname{sgn}(c\eta_i)\big(\operatorname{sgn} O_b(t_i), -\operatorname{sgn} E_b(t_i)\big), \quad \operatorname{sgn}\eta_i = (-1)^{m-i}. \tag{5.4}$$

The numerator interval indices q_i are nondecreasing, and the denominator signs must fit into a nondecreasing sequence of regions from Table 1. If $t_1 = 0$, equation (3.4) gives $b_5 = 0$ and

$$\mu_1 a_{1,5} = c\eta_1 b_4 > 0.$$

Thus $c\eta_1 > 0$, and the interval indices of all remaining positive nodes are at least 1.

The enumeration starts with the alternatives in Proposition 4.7. For $m = 5$, the numerator is quartic and $c > 0$. For $m = 6$, the numerator may be quartic or quintic, and either sign of c must initially be allowed. For each choice, include both the possibility of an initial zero node and its absence. The condition $c\eta_1 > 0$ removes the incompatible sign when zero is present.

Table 2: Remaining contact configurations and denominator signs at the nodes

Case	Intervals, vertices, signs	Case	Intervals, vertices, signs
Q1	$(0,1,1,2,3,4)$	Q4	$(0,2,2,3,3,4)$
	$DAABCD$		$DBBCCD$
	$-+,\ ++,\ --,\ -+,\ ++,\ +-$		$-+,\ -+,\ +-,\ --,\ ++,\ +-$
Q2	$(0,1,1,3,3,4)$	Q5	$(1,2,2,3,3,4)$
	$DAACCD$		$ABBCCD$
	$-+,\ ++,\ --,\ --,\ ++,\ +-$		$--,\ -+,\ +-,\ --,\ ++,\ +-$
Q3	$(0,1,2,3,3,4)$		
	$DABCCD$		
	$-+,\ ++,\ +-,\ --,\ ++,\ +-$		

Each positive node then lies in one of four or five ordered numerator intervals, and (5.4) fixes its denominator signs.

These choices give a finite enumeration with a precise rejection rule. Append the numerator interval indices in nondecreasing order. For each new node, choose the smallest denominator region compatible with its required signs and not earlier than the region chosen at the preceding node. If no such region exists, discard that prefix. This choice loses no compatible sequence: by induction, its chosen region is no later than the corresponding region in any admissible assignment. Conversely, a completed sequence of these choices is a compatible assignment. Thus the rule tests exactly the region-order constraint, rather than an additional sufficient condition for contact realizability. The sign-code formula and full recurrence are given in Appendix A.

The resulting candidate list has 26 rows: two with $m = 5$, three with $m = 6$ and a quartic numerator, and twenty-one with $m = 6$ and a quintic numerator. Every actual representation occurs in this list, since its numerator indices are ordered, its signs satisfy (5.4), and its denominator regions are ordered by Proposition 2.4. Multiplicity then removes twelve rows having three contacts at one vertex. Lemma 5.3 removes six rows with two A contacts followed by two B contacts. The two remaining quartic rows with an initial zero node violate Lemma 5.2(b); the remaining quintic row with an initial zero node violates Lemma 5.4. The five surviving rows are exactly Q1–Q5 in Table 2. Table 3 records each candidate and the exclusion applied to it. This step uses the stated finite enumeration; region order alone does not establish the realizability of any surviving row.

The surviving rows all have $c > 0$, all nodes positive, and $b_0 > 0$. If $b_5 = 0$, then $I_0 = \varnothing$, so Q1–Q4 are impossible, while Lemma 5.4 excludes Q5. Hence $b_5 > 0$, giving

$$0 < e < k < f < \ell. \tag{5.5}$$

Finally, the denominator signs in Table 2 follow directly from (5.4), and (5.1) ensures that they are the signs of the corresponding contact vertices. □

6 Root ratio comparisons

The five configurations contain repeated contacts at vertices sharing one component. The two contacts determine square factors of the pairing polynomials, while the shared component makes their values equal at the roots of the other numerator component. These two expressions for a value ratio give incompatible strict bounds.

Retain the stable family K and its nonnegative pairings P_J. Throughout this section, $b_0, b_5 > 0$ and the numerator roots satisfy (5.5). If a vertex J has contacts at two positive

points $r < s$, then (5.2) and $P_J(0) = a_{J,5}b_5 > 0$ give $d_J > 0$. Retaining the original leading coefficient b_0, write

$$\frac{P_J(t)}{b_0} = (t + \delta_J)(t - r)^2(t - s)^2, \quad \delta_J = \frac{d_J}{b_0} > 0. \tag{6.1}$$

Lemma 6.1 (Strict bounds for a shared value ratio)**.** Suppose two contact vertices J_1, J_2 have the form (6.1) and take equal positive values at each of two points $0 < x < y$, that is, $P_{J_1}(x) = P_{J_2}(x) > 0$, $P_{J_1}(y) = P_{J_2}(y) > 0$. Write $P_{J_j}(t)/b_0 = (t + \delta_{J_j})H_j(t)^2$ for $j = 1, 2$, where H_j is the monic quadratic factor determined by the two contacts of J_j. Put

$$\rho_1 = \frac{|H_1(x)|}{|H_1(y)|}, \quad \rho_2 = \frac{|H_2(x)|}{|H_2(y)|}, \quad R = \frac{P_{J_1}(x)}{P_{J_1}(y)}.$$

Then

$$\frac{x}{y}\rho_1\rho_2 < R < \rho_1\rho_2. \tag{6.2}$$

Proof. Write $\delta_j = \delta_{J_j}$, and put $X = (x+\delta_1)/(y+\delta_1)$ and $Y = (x+\delta_2)/(y+\delta_2)$. Since x, y, δ_1, δ_2 are strictly positive and $x < y$, $x/y < X, Y < 1$. The two factorizations give $R = X\rho_1^2 = Y\rho_2^2$, with $\rho_1, \rho_2 > 0$. Hence $R^2 = XY\rho_1^2\rho_2^2$. Taking the positive square root gives $R = \sqrt{XY}\rho_1\rho_2$, proving (6.2). □

For $x < y$, the functions

$$h(z) = \frac{z - x}{y - z}, \quad g(z) = \frac{z - x}{z - y}, \quad h_0(z) = \frac{x - z}{y - z}$$

are, respectively, strictly increasing on (x, y), strictly decreasing on (y, ∞), and strictly decreasing on $(-\infty, x)$. Also $g(z) > 1$ for $z > y$.

Proposition 6.2 (Exclusion for vertices B and C)**.** There are no contacts r, s of B and u, v of C satisfying $k < r < s < f < u < v < \ell$ and

$$O_+(r) > 0, \quad O_+(s) < 0, \quad O_+(u) < 0, \quad O_+(v) > 0.$$

In particular, configurations Q4 and Q5 are impossible.

Proof. The common odd component of B, C is O_+. Let its two simple positive roots be $\alpha < \beta$. A quadratic with positive leading coefficient is negative between its roots, so $r < \alpha < s$ and $u < \beta < v$. The full strict ordering is

$$0 < e < k < r < \alpha < s < f < u < \beta < v < \ell. \tag{6.3}$$

Since E_b vanishes at e, f,

$$P_B(t) = P_C(t) = tO_b(t)O_+(t) \qquad (t = e, f).$$

The strict ordering excludes coincidences between the evaluation points and the contacts, so Lemma 6.1 applies. Define the positive quantities

$$\rho_1 = \frac{(r - e)(s - e)}{(f - r)(f - s)}, \quad \rho_2 = \frac{(u - e)(v - e)}{(u - f)(v - f)},$$
$$Z = \frac{k - e}{f - k}\frac{\alpha - e}{f - \alpha}\frac{\ell - e}{\ell - f}\frac{\beta - e}{\beta - f}.$$

Here ρ_1, ρ_2 are the absolute value ratios of the contact quadratics. Since $O_b = b_0(t-k)(t-\ell)$ and $O_+ = (t-\alpha)(t-\beta)$, all scale factors cancel, giving

$$R = \frac{P_B(e)}{P_B(f)} = \frac{e}{f} Z. \tag{6.4}$$

Apply h, g with $x = e, y = f$. By (6.3),

$$h(k) < h(r), \quad h(\alpha) < h(s), \quad g(\beta) < g(u), \quad g(\ell) < g(v).$$

All factors are strictly positive, so $Z < \rho_1\rho_2$. Equation (6.4) gives $R < (e/f)\rho_1\rho_2$, whereas Lemma 6.1 gives $R > (e/f)\rho_1\rho_2$, a contradiction. The vertex order and signs in Q4 and Q5 satisfy the stated hypotheses. □

Proposition 6.3 (Exclusion for vertices A and D)**.** There are no contacts r, s of A and w, z of D satisfying $0 < w < e < r < s < k < f < \ell < z$ and

$$O_-(r) > 0, \quad O_-(s) < 0, \quad O_-(z) < 0.$$

In particular, configurations Q1 and Q2 are impossible.

Proof. The common odd component of A, D is O_-. Its roots $\alpha < \beta$ therefore satisfy $r < \alpha < s$ and $z < \beta$: both s, z lie between the roots, while $r < s$ and positivity at r place r before the first root. The full ordering is

$$0 < w < e < r < \alpha < s < k < f < \ell < z < \beta. \tag{6.5}$$

Since $E_b(e) = E_b(f) = 0$, we have $P_A(t) = P_D(t) = tO_b(t)O_-(t)$ at $t = e, f$. Define

$$\rho_1 = \frac{(r-e)(s-e)}{(f-r)(f-s)}, \quad \rho_2 = \frac{(e-w)(z-e)}{(f-w)(z-f)},$$
$$Z = \frac{k-e}{f-k}\frac{\alpha-e}{f-\alpha}\frac{\ell-e}{\ell-f}\frac{\beta-e}{\beta-f}.$$

Put $R = P_A(e)/P_A(f)$ and take $x = e, y = f$. The factorization of the common odd component gives $R = (e/f)Z$, while Lemma 6.1 gives $R < \rho_1\rho_2$. The ordering (6.5) and monotonicity give

$$h(k) > h(s), \quad h(\alpha) > h(r), \quad g(\ell) > g(z), \quad g(\beta) > 1.$$

Therefore,

$$Z = h(k)h(\alpha)g(\ell)g(\beta) > \rho_1 \frac{z-e}{z-f} = \rho_1\rho_2 \frac{f-w}{e-w}.$$

Since $w > 0$,

$$\frac{f-w}{e-w} - \frac{f}{e} = \frac{w(f-e)}{e(e-w)} > 0.$$

It follows that $R = (e/f)Z > \rho_1\rho_2$, a contradiction. The hypotheses hold for both Q1 and Q2 by Table 2. □

Proposition 6.4 (Exclusion for vertices C and D)**.** There are no contacts u, v of C and w, z of D satisfying $0 < w < e < k < f < u < v < \ell < z$ and

$$E_-(w) < 0, \quad E_-(u) < 0, \quad E_-(v) > 0.$$

In particular, configuration Q3 is impossible.

Proof. The common even component of C, D is E_-. If its two simple positive roots are $\alpha < \beta$, then $\alpha < w$ and $u < \beta < v$. Thus

$$0 < \alpha < w < e < k < f < u < \beta < v < \ell < z. \tag{6.6}$$

This time take $x = k, y = \ell$. Since $O_b(k) = O_b(\ell) = 0$,

$$P_C(t) = P_D(t) = E_b(t)E_-(t) \quad (t = k, \ell).$$

Define

$$\rho_1 = \frac{(u-k)(v-k)}{(\ell-u)(\ell-v)}, \quad \rho_2 = \frac{(k-w)(z-k)}{(\ell-w)(z-\ell)},$$
$$Z = \frac{k-e}{\ell-e}\frac{f-k}{\ell-f}\frac{k-\alpha}{\ell-\alpha}\frac{\beta-k}{\ell-\beta}.$$

The common scale factor in E_bE_- is $b_1 l_1 > 0$ and cancels in the value ratio. Hence

$$R = \frac{P_C(k)}{P_C(\ell)} = Z. \tag{6.7}$$

Using $f < u$, $\beta < v$, $e > w$, and $\alpha > 0$, the corresponding monotonicity properties give

$$\frac{f-k}{\ell-f}\frac{\beta-k}{\ell-\beta} < \rho_1,$$
$$\frac{k-e}{\ell-e} < \frac{k-w}{\ell-w},$$
$$\frac{k-\alpha}{\ell-\alpha} < \frac{k}{\ell}.$$

Since $(z-k)/(z-\ell) > 1$, we obtain

$$R = Z < \frac{k}{\ell}\rho_1\frac{k-w}{\ell-w} < \frac{k}{\ell}\rho_1\rho_2.$$

This contradicts the bound $R > (k/\ell)\rho_1\rho_2$ in Lemma 6.1. Configuration Q3 has precisely the required vertex order and component signs. □

7 Proof of the main theorem

Proof of Theorem 2.1. Suppose a stable family K has no common real numerator of degree five. Lemma 3.2 embeds it in a stable interval family with positive interval lengths, which likewise has no such numerator. Proposition 3.3 then supplies a critical family, a nonzero polynomial b with nonnegative pairings throughout that family, and a finite weighted representation. Propositions 4.4 and 4.6 give coprime numerator components and the constant interpolation relation; Proposition 4.7 gives the two possible support sizes. Proposition 5.5 leaves only Q1–Q5.

Proposition 6.2 excludes Q4 and Q5, Proposition 6.3 excludes Q1 and Q2, and Proposition 6.4 excludes Q3. This contradiction proves that a single $b \in \mathbb{R}[s]$ of actual degree five satisfies (2.2) for every $a \in K$ and every $\omega \in \mathbb{R}$. □

The quintic assumption enters at two related steps. First, the numerator has six coefficients, and the critical vector determines a five-dimensional hyperplane, so at most six support points are needed. Second, both components have degree at most two. The moment identities and coprimality therefore reduce the interpolation identity to a constant proportionality relation. These restrictions turn the constraints over infinitely many denominators and frequencies into the exclusion of finitely many configurations. For higher-order interval families, the corresponding degree comparisons do not directly yield the same restrictions, and no higher-order extension is asserted here.

8 Formal verification

The proof has been fully formalized in Lean 4. The source code is available in the public repository `dgjszy/quintic-spr-proof`. The main entry point is the theorem `SPR.N5.Direct.robustSPR` in `Lean/Main.lean`. Its statement matches Theorem 2.1: a single real polynomial b of degree five satisfies $\mathrm{Re}(b(i\omega)/a(i\omega)) > 0$ for every $a \in K$ and every $\omega \in \mathbb{R}$.

The formal development follows the mathematical structure of this paper. It first establishes the imaginary-axis decomposition, Hermite–Biehler interlacing, and the uniform root bands. It then formalizes the critical-family construction and finite convex certificate, the moment-interpolation identity, the contact classification, and the three root-ratio exclusions. These results are combined to rule out every weak critical certificate and derive the strict synthesis theorem. The Lean code is maintained separately from the manuscript, so the paper remains readable as an independent mathematical proof while the repository supplies a machine-checked verification of the entire argument.

A Complete classification of contact configurations

We give the finite classification used in Proposition 5.5 and its recurrence rule. The classification records only two kinds of information: the numerator interval containing each node, and the denominator component signs at that node. The first is represented by the interval index q_i from Section 5; the second is encoded below by an integer. This encoding serves only to list a finite table and introduces no new contact conditions. Encode $++, -+, --, +-$ by $0, 1, 2, 3$, respectively. The allowed ordered regions for each nonzero denominator sign pair are

$$Z(0) = \{0, 1, 7, 8\}, \quad Z(1) = \{1, 2, 3\}, \quad Z(2) = \{3, 4, 5\}, \quad Z(3) = \{5, 6, 7\}.$$

Given m, the numerator degree, and the sign of c, put $\nu = \mathbf{1}_{\{c<0\}}$ and $\varepsilon = \mathbf{1}_{\{t_1=0\}}$. Set $h = 5$ in the quintic case and $h = 4$ in the quartic case. Here ε records only whether a zero node is included in the support; $b_5 = 0$ does not imply $\varepsilon = 1$. Thus branches without a zero node must initially allow $e = 0$ as well. The interval indices of every actual positive-contact sequence satisfy

$$\varepsilon \le q_{1+\varepsilon} \le \cdots \le q_m \le h - 1.$$

We enumerate these nondecreasing integer sequences using only necessary conditions. In particular, when $\varepsilon = 0$, we do not yet remove sequences containing index 0 because of a possible $e = 0$; these empty-interval cases are excluded separately at the end of Proposition 5.5. In numerator interval q_i, the sign pairs cycle through $++, -+, --, +-$. Equation (5.4) first sends the signs of (E_b, O_b) to those of $(O_b, -E_b)$, which adds three to the code. Multiplication by a negative sign adds two. Combining this with $\operatorname{sgn} \eta_i = (-1)^{m-i}$ gives the denominator sign code at every positive node:

$$\sigma_i \equiv q_i + 3 + 2(m - i + \nu) \pmod 4, \quad 0 \le \sigma_i \le 3. \tag{A.1}$$

If a zero node is present, we also require $m - 1 + \nu$ to be even and set $\sigma_1 = 0$.

For each such sequence, start with $g_0 = 0$ and define recursively

$$g_i = \min\{z \in Z(\sigma_i) : z \ge g_{i-1}\}. \tag{A.2}$$

Discard the sequence if the set is empty at any step. This test is equivalent to the existence of nondecreasing region indices $\zeta_i \in Z(\sigma_i)$. If the recurrence succeeds, take $\zeta_i = g_i$. Conversely, if such indices ζ_i exist, induction gives $g_i \le \zeta_i$, so every step of the recurrence is possible. This proves that the recurrence is both necessary and sufficient for region compatibility.

For example, take $m = 6$, $c > 0$, and no zero node. For the Q1 sequence $q = (0, 1, 1, 2, 3, 4)$, equation (A.1) gives $\sigma = (1, 0, 2, 1, 0, 3)$, and (A.2) gives $g = (1, 1, 3, 3, 7, 7)$, so the sequence is retained. In contrast, the first three sign codes for $q = (0, 0, 0, 0, 0, 0)$ are $(1, 3, 1)$. The recurrence gives $g_1 = 1$ and $g_2 = 5$, then fails at the third step because $Z(1) = \{1, 2, 3\}$ contains no element at least 5.

The full enumeration can be specified recursively without choosing any frequency values. Loop over

$$(m, \deg b, \nu) \in \{(5, 4, 0), (6, 4, 0), (6, 4, 1), (6, 5, 0), (6, 5, 1)\}$$

and $\varepsilon \in \{0, 1\}$, rejecting $\varepsilon = 1$ when $m - 1 + \nu$ is odd. For an empty positive-node prefix, allow the next index to range from ε to $h - 1$; for a nonempty prefix, allow it to range from the last chosen index to $h - 1$. After appending an index, compute its sign by (A.1) and update (A.2). A failed update ends that branch. Retain a branch when it contains $m - \varepsilon$ positive-node indices. With a zero node, initialize the region at $g_1 = 0$; otherwise initialize at $g_0 = 0$.

Every branch has at most six extensions, each with at most five choices, so the procedure is finite. Induction on prefix length generates every nondecreasing index sequence. The induction following (A.2) shows that pruning removes exactly those prefixes with no compatible region assignment. It follows that Table 3 is the complete output for these necessary conditions. It contains every actual contact configuration, without asserting that every row is realizable. The 26 rows split into 12 marked N, 6 marked AB, 2 marked Z4, 1 marked Z5, and the 5 retained rows Q1–Q5.

Read each table down the left half, then down the right half. In Table 3, the triple $(m, \deg b, \operatorname{sgn} c)$ records the number of finite nodes, the numerator degree, and the sign of c. When $m = 5$, there is also a support point at infinity. Commas are omitted from interval sequences, and $*$ denotes an additional initial zero node. The code N denotes three positive contacts at one vertex; Z4 denotes two positive contacts at one vertex when $b_0 = b_5 = 0$. The codes AB and Z5 refer to Lemmas 5.3 and 5.4, respectively. Q1–Q5 are the five configurations retained in the main text.

Table 3: All compatible interval sequences and their exclusions

$(m, \deg b, \operatorname{sgn} c)$	Sequence and outcome	$(m, \deg b, \operatorname{sgn} c)$	Sequence and outcome
$(5, 4, +)$	11223 (AB)	$(6, 5, +)$	011234 (Q1)
$(5, 4, +)$	$*1223$ (Z4)	$(6, 5, +)$	011244 (N)
$(6, 4, +)$	011223 (AB)	$(6, 5, +)$	011334 (Q2)
$(6, 4, -)$	112233 (AB)	$(6, 5, +)$	011344 (N)
$(6, 4, -)$	$*12233$ (Z4)	$(6, 5, +)$	012334 (Q3)
$(6, 5, +)$	001124 (N)	$(6, 5, +)$	013334 (N)
$(6, 5, +)$	001134 (N)	$(6, 5, +)$	013344 (N)
$(6, 5, +)$	001334 (N)	$(6, 5, +)$	022334 (Q4)
$(6, 5, +)$	002334 (N)	$(6, 5, +)$	023334 (N)
$(6, 5, +)$	011124 (N)	$(6, 5, +)$	023344 (N)
$(6, 5, +)$	011134 (N)	$(6, 5, +)$	122334 (Q5)
$(6, 5, +)$	011223 (AB)	$(6, 5, -)$	112233 (AB)
$(6, 5, +)$	011224 (AB)	$(6, 5, -)$	$*12233$ (Z5)

For example, the first, second, and sixth nodes of 001124 all correspond to vertex D, so N excludes this sequence. The sequence 11223 has two A contacts followed by two B contacts, so AB excludes it. The quartic cases with a zero node have the multiplicity forbidden by Z4, while the quintic case $*12233$ with a zero node has the ordering forbidden by Z5. Every exclusion in the table therefore follows from a polynomial identity or degree comparison proved in the main text.

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